



A MULTI-SITE NETWORK w. ADDRESS TRANSLATION

1. Introduction

1.1. Objectives

This work aims to illustrate the address translation in a multi-site network.

1.2. Outcomes

By completing this work through step-by-step guideline, students will be able to understand:

- Configuration Steps for NAT Address Translation: show the step-by-step process for configuring Network Address Translation (NAT) on a router, including defining inside and outside interfaces, specifying NAT rules, and applying appropriate access control lists (ACLs).
- Illustration of Network Access to a Server Behind NAT: shows how to access a server located behind a NAT device, including the use of static NAT to ensure that incoming requests are correctly routed to the server.
- Examination of Packet Address Translation: shows how packet address translation works during data transmission, including the translation of source and destination IP addresses and ports, and the implications for both internal and external communications.

2. Testbest setup

In this section, we develop a network system with 2 sites at LTK and DiAn. The sites are connected through interconnected routers provided by ISP.



2.1. Network devices

Due to the geographical separation, each site has an internal network (access network). Each internal network forms a subnet (or access network) with a gateway router. The connections among gateway routers are provided by ISP with public IP addresses.

In each site, the hosts will serve as the end device, capable of sending and receiving data while the switches broadcastly connect these hosts together. The local IP addresses are managed by a DHCP server. To demonstrate the remote (external) access connection, we equip each subnet with a web HTTP server.

The router will connect the subnet to external networks and act as a gateway (router) to translate the local IP address to a limited set (a pool) of public IP addresses. To keep the demo simple, we assume each gateway has only one public IP address.

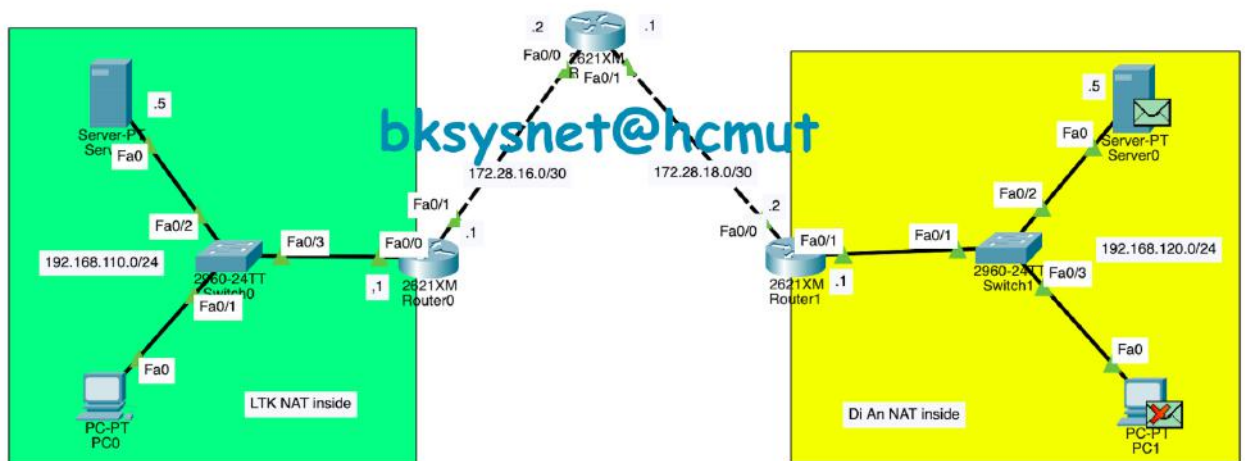


Figure 2.1 Two sites with inter-connected router



2.2. Device configuration

2.2.1. Configure server with a static IP address and DHCP, HTTP service

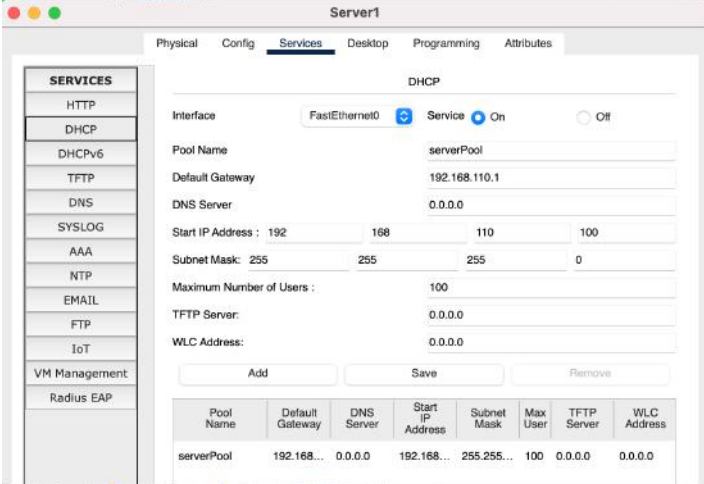
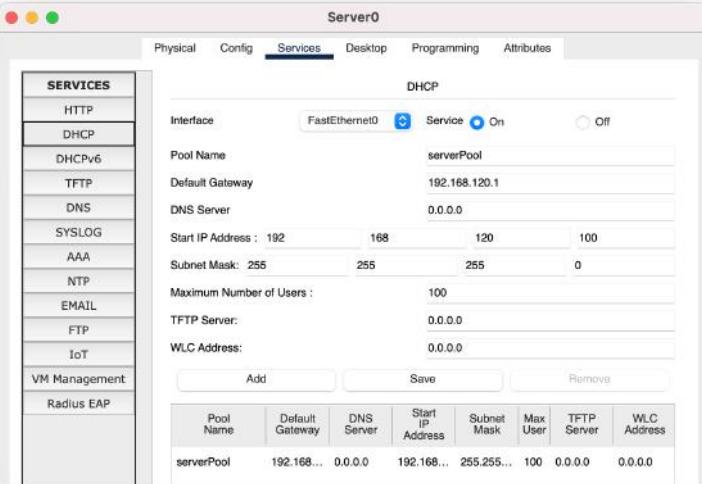
On each server, we enable HTTP and HTTPS service.

Setup the static IP addresses.

Site LTK	Site DiAn
Server 1 IP : 192.168.110.5 Server 1 netmask: 255.255.255.0 Server 1 default gateway: 192.168.110.1	Server 0 IP : 192.168.120.5 Server 0 netmask: 255.255.255.0 Server 0 default gateway: 192.168.120.1

Setup the DHCP service

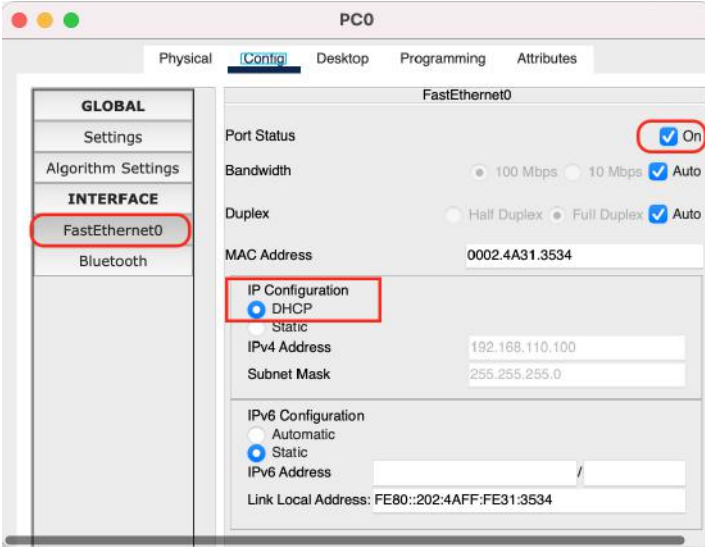
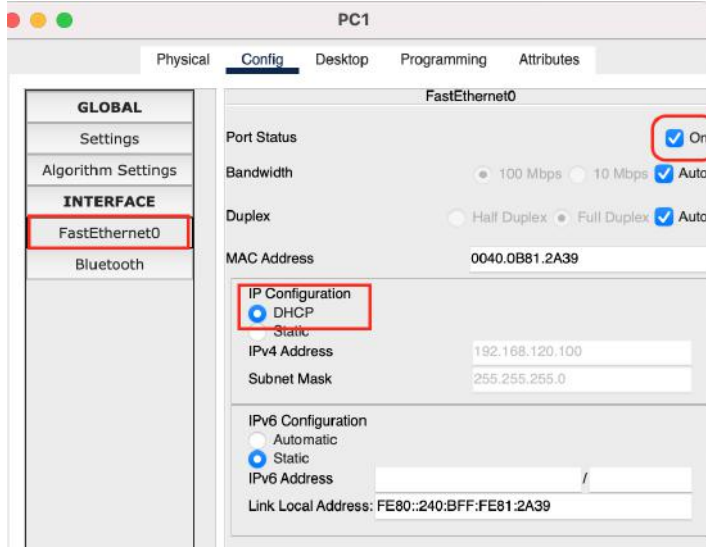
Site LTK	Site DiAn
Server 1 Interface : FastEthernet0 Server 1 Pool Name: serverPool Server 1 Default gateway: 192.168.110.1 Server 1 Start IP Address: 192.168.110.100 Server 1 Subnet Mask: 255.255.255.0 Server 1 Max No User: 100	Server 0 Interface : FastEthernet0 Server 0 Pool Name: serverPool Server 0 Default gateway: 192.168.110.1 Server 0 Start IP Address: 192.168.120.100 Server 0 Subnet Mask: 255.255.255.0 Server 0 Max No User: 100

	
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2.2.2. Configure PCs in DHCP client mode

On each PC, we enable DHCP mode in Config>Interface>FastEthernet

PC 0	PC 1
	

2.2.3. Configure inter-connected routers

Configure Gateway Router Site LTK

```
Would you like to enter the initial configuration dialog? [yes/no]: no
Router>en
Router#configure terminal
Router(config)#interface FastEthernet 0/0
Router(config-if)#ip address 192.168.110.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config)#interface FastEthernet 0/1
Router(config-if)#ip address 172.28.16.1 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#access-list 1 permit 172.28.16.0 0.0.0.3
Router(config)#access-list 1 permit 192.168.110.0 0.0.0.255

Router(config)#ip nat inside source list 1 interface FastEthernet 0/1 overload
Router(config)#ip nat inside source static tcp 192.168.110.5 80 172.28.16.1 1111
```



```
Router(config)#ip nat outside source static tcp 172.28.16.1 1111 192.168.110.5 80

Router(config)#router ospf 6000
Router(config-router)#network 172.28.16.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#exit
Router#wr
Building configuration...
[OK]
```

Configure Gateway Router at site DiAn

```
Would you like to enter the initial configuration dialog? [yes/no]: no
Router>en
Router#configure terminal
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.120.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config)#interface FastEthernet0/0
Router(config-if)# ip address 172.28.18.2 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#access-list 1 permit 172.28.18.0 0.0.0.3
Router(config)#access-list 1 permit 192.168.120.0 0.0.0.255

Router(config)#ip nat inside source list 1 interface FastEthernet 0/0 overload
Router(config)#ip nat outside source static tcp 172.28.18.2 2222 192.168.120.5 80
Router(config)#ip nat inside source static tcp 192.168.120.5 80 172.28.18.2 2222

Router(config)#router ospf 6000
Router(config-router)#network 172.28.18.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#exit
Router#wr
Building configuration...
[OK]
```

Configure Inter-connected routers

```
Would you like to enter the initial configuration dialog? [yes/no]: no
Router>en
Router#configure terminal
Router(config)#interface FastEthernet0/0
```



```
Router(config-if)#ip address 172.28.16.2 255.255.255.252
Router(config-if)#no shutdown

Router(config)#interface FastEthernet0/1
Router(config-if)# ip address 172.28.18.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#router ospf 6000
Router(config-router)#network 172.28.16.0 0.0.0.3 area 0
Router(config-router)#network 172.28.18.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#exit
Router#wr
Building configuration...
[OK]
```

2.2.4. Verify address translation.

Gateway router at site LTK:

```
Router>en
Router#show ip nat statistics
Total translations: 1 (1 static, 0 dynamic, 0 extended)
Outside Interfaces: FastEthernet0/1
Inside Interfaces: FastEthernet0/0
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:

outer#show access-lists 1
Standard IP access list 1
    permit 172.28.16.0 0.0.0.3
    permit 192.168.110.0 0.0.0.255

Router#show ip nat translations
Pro  Inside global      Inside local    Outside local   Outside global
tcp  172.28.16.1:1111     192.168.110.5:80 ---             ---
tcp  ---                ---            192.168.110.5:80 172.28.16.1:1111
```

In each gateway router, verify the inside and outside interfaces, access list and static NAT address translations.



```
Router>en
Router#show ip nat statistics
Total translations: 1 (1 static, 0 dynamic, 0 extended)
Outside Interfaces: FastEthernet0/0
Inside Interfaces: FastEthernet0/1
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:

Router#show access-lists 1
Standard IP access list 1
    permit 172.28.18.0 0.0.0.3
    permit 192.168.120.0 0.0.0.255

Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  172.28.18.2:2222    192.168.120.5:80  ---               ---
tcp  ---                ---              192.168.120.5:80  172.28.18.2:2222
```

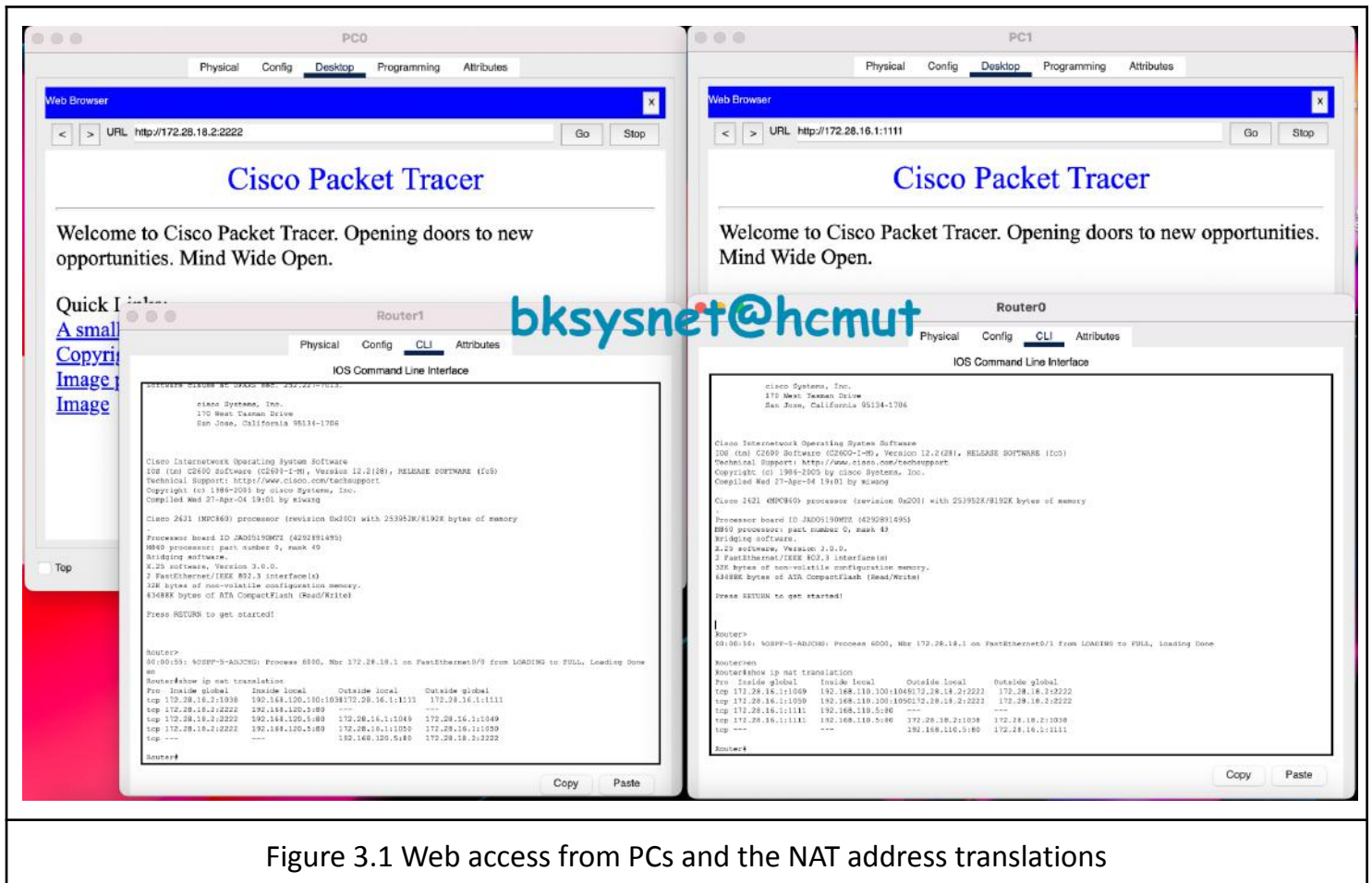
In each gateway router, verify the inside and outside interfaces, access list and static NAT address translations.

3. Packet Tracing

In this section, we perform external access through address mapping into the site's internal web server. On each PC, we access through the gateway router public IP address with static NAT port mapping. In each access, the NAT service will automatically translate between the inside and outside through an entry in the NAT address translation table.

[Figure 3.1](#) shows the successful web browser accesses and the content of NAT address translation table.

In the following sections, we perform a tracing of a packet sending from PC0 at the site LTK to the server at the site DiAn. We show the address translation progress through each NAT inside to the NAT outside.



3.1. Site LTK

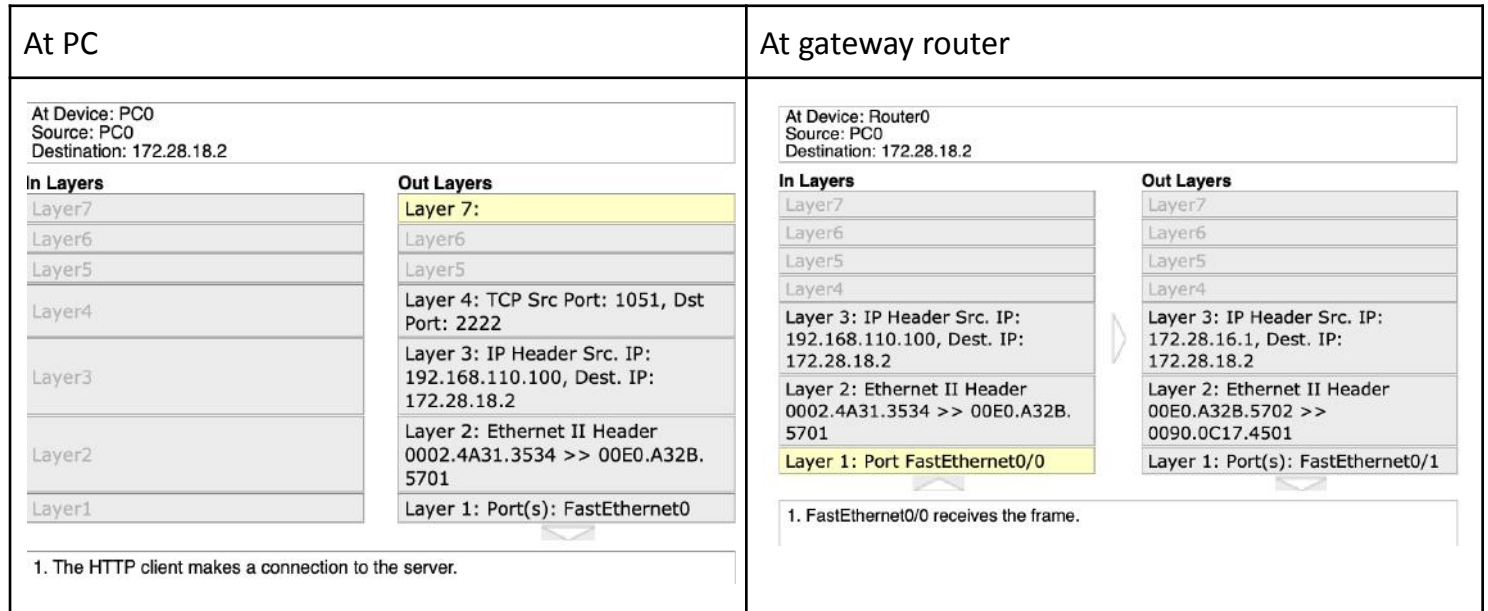
The NAT address translation table

```
Router>en
Router#show ip nat translation
```

Pro	Inside global	Inside local	Outside local	Outside global
tcp	172.28.16.1:1049	192.168.110.100:1049	172.28.18.2:2222	172.28.18.2:2222
tcp	172.28.16.1:1050	192.168.110.100:1050	172.28.18.2:2222	172.28.18.2:2222
tcp	172.28.16.1:1111	192.168.110.5:80	---	---
tcp	172.28.16.1:1111	192.168.110.5:80	172.28.18.2:1038	172.28.18.2:1038
tcp	---	---	192.168.110.5:80	172.28.16.1:1111

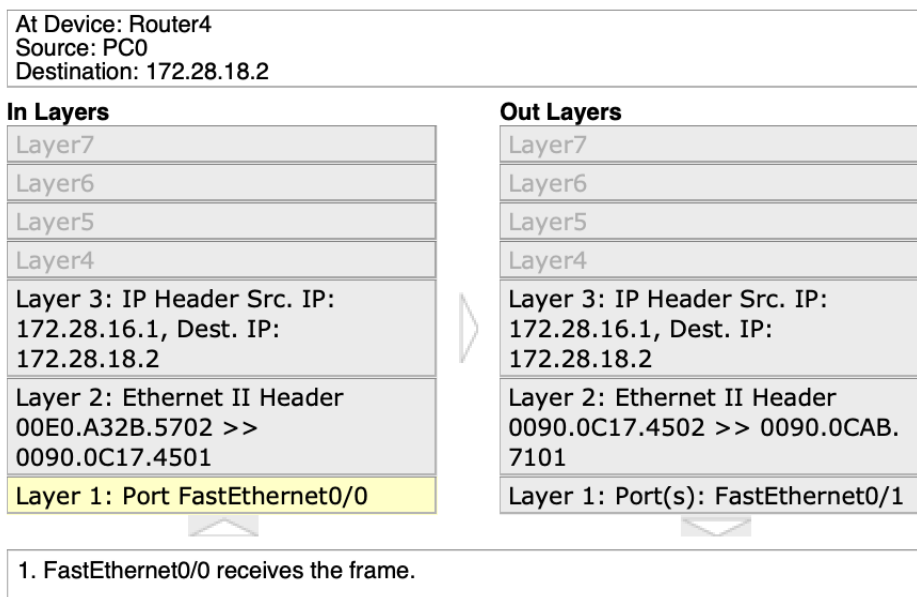


The packet address is translated the source addresses from inside 192.168.110.100 to 172.28.16.1



3.2. At the interconnected router

The address is public addresses of 172.28.16.1 and 172.28.18.2



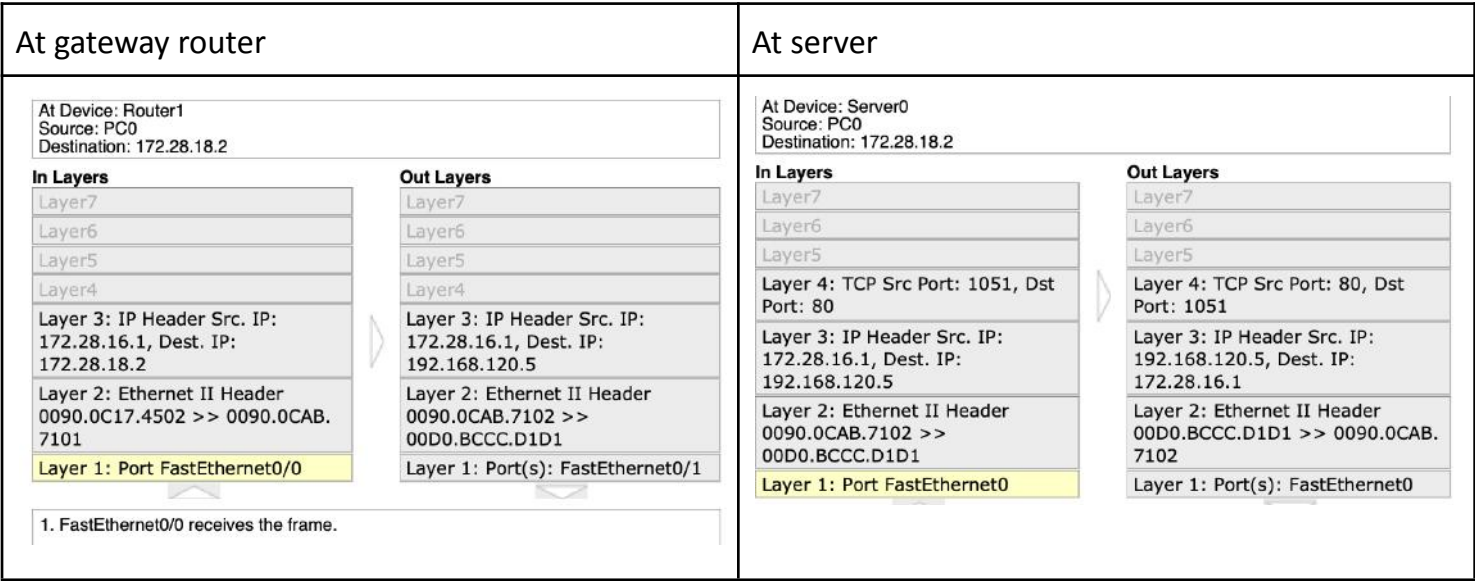


3.3. Site DiAn

The NAT address translation table

Router#show ip nat translation				
Pro	Inside global	Inside local	Outside local	Outside global
tcp	172.28.18.2:1038	192.168.120.100:1038	172.28.16.1:1111	172.28.16.1:1111
tcp	172.28.18.2:2222	192.168.120.5:80	---	---
tcp	172.28.18.2:2222	192.168.120.5:80	172.28.16.1:1049	172.28.16.1:1049
tcp	172.28.18.2:2222	192.168.120.5:80	172.28.16.1:1050	172.28.16.1:1050
tcp	---	---	192.168.120.5:80	172.28.18.2:2222

The packet address is translated the destination addresses from inside 172.28.18.2 to 192.168.120.5



4. Conclusions

In this work, we perform the address translation between the inside NAT address and outside NAT address. We verify the address translation at each gateway router of both sites LTK and DiAn. The address translation is examined at each hop associated with the NAT address translation table entries. This packet tracing helps illustrate the NAT address translation in the course CO3093 and CO3094 at HCMUT.